Visual Analytics of Motor Vehicle Accidents

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***Abstract*—In this paper, we present a visual analytics solution to analyze and visualize motor vehicle accidents in metropolitan cities. Through advanced visualization techniques and overlaying accident severity data onto citywide maps, the research provides valuable insights for city officials. These insights support targeted interventions, contributing to an overall enhancement of urban safety. In contrast to conventional visualizations, this research offers a unique perspective crucial for informed policymaking and the development of effective intervention strategies. Evaluation results on real-world data in the Canadian metropolitan city of Toronto show the practicality of our solution in revealing accident patterns in the city. The discovered information and knowledge helps decision makers to improve city services.**

Keywords—information visualization, visual analytics, data science, visualization, accidents, alcohol-related, resource distribution, time

I. Introduction

In the current era of big data, data are everywhere. Embedded in these data are implicit, previously unknown and useful knowledge that can be discovered by big data science [1]—which makes good use of data mining [2-7], machine learning [8], statistics [9], and visualization [10, 11]. The latter has been applied to many real-world application areas like transportation [12], music [13], biomedicine [14-18], social networks [19, 20], uncertain data [21], and time series [22, 23]. This paper focuses on transportation analytics.

To elaborate, in any bustling metropolis traffic accidents occur with regularity, posing a continual challenge for city officials tasked with resource allocation. For instance, Toronto, a Canadian city with 158 neighborhoods, presents a unique challenge for managing these incidents. We aim to provide a comprehensive solution to this complex challenge and support city officials in optimizing resource distribution. Here, our key focus lies in seamlessly integrating accident data with specific location information. By doing so, we strive to empower city authorities to pinpoint areas that demand concentrated attention and strategic intervention.

To enhance the utility of our findings, we combine traditional charts and graphs with our novel mapping technique, presenting the data in an accessible format for city officials. This not only simplifies the interpretation of complex patterns but also aids in fostering a deeper understanding of the accident dynamics within different areas of the city. Central to our methodology is the ability for the city's resources to be used efficiently. We acknowledge the need for efficient resource allocation, directing attention to areas where the impact of interventions is likely to be most significant. This model serves as a scalable framework that can be extended to any city grappling with similar challenges, provided they have comparable accident data and neighborhood profiles. In essence, our initiative represents a holistic approach to urban safety management, leveraging data analysis and visualization to offer actionable insights that can significantly improve the city officials' decision-making processes.

Key contributions of this paper include our design and development of a visual analytics solution to analyze and visualize motor vehicle accidents. The remainder of this paper is organized as follows. The next section provides background and related work. Section III describes our visual analytics solution. Section IV shows evaluation results. Finally, Section V draws conclusions.

II. Background and Related Work

Numerous studies [24-30] have delved into the analysis of accidents; however, critical observation reveals significant limitations in their methodologies, particularly with respect to visualization. A considerable portion of these studies either entirely omit visualizations [24-26] or neglect the integration of maps into their analysis [27-30]. Furthermore, among those studies that do incorporate maps, there is a prevalent deficiency in detail necessary for comprehensive visualization [31, 32]. These limitations pose substantial challenges, especially for city planners who are crucial stakeholders in deciphering and leveraging accident data for urban planning. Previous studies [24-26] often present data in tabular formats, which may impede comprehension for city planners unaccustomed to interpreting raw data tables. Alternatively, an excess of charts can overwhelm planners, hindering their ability to extract actionable insights.

Even in instances where maps are included, the prevalent issue lies in the lack of finer detail. Maps typically restrict themselves to broad district-level shading or identify large hotspots, overlooking the granularity required for identifying specific problematic areas such as individual intersections that demand improvement. This oversight poses a considerable challenge for city planners seeking targeted solutions for localized issues within their jurisdiction. As such, there exists a pressing need for studies that not only employ visualizations but also prioritize a finer level of detail in their mapping approaches to better serve the practical needs of urban planners.

Our key contributions of this paper come in two-fold. From visualization aspects, this paper revolves around one primary objective: visualization techniques for actionable insights. We achieve this by:

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- *mapping accidents* through leveraging the coordinates of each accident to create detailed maps, allowing for a spatial representation of accidents;
- incorporating the *frequency and recency* of accidents in specific locations, which provides a cleaner view of the accident landscape, enabling a more detailed understanding; and
- presenting a new visualization technique—*gradient-dot heatmaps*, which employs gradients to dots to show the recency of the accident. Faded dots indicate a non-recent accident and can be interpreted as resolved issues. A full-color dot signifies a recent issue and could be interpreted as an ongoing problem, which needs to be addressed. This intuitive representation aids in the quick identification of areas requiring immediate attention

In terms of visual analytics and socio-economic impact, our combined efforts aim to offer city officials a comprehensive and actionable overview of the accident landscape. The detailed maps and insightful visualizations empower decision-makers to implement targeted interventions and policies, improving safety measures in high-risk areas. Ultimately, our contributions contribute to the creation of a safer urban environment in Toronto. Our process is split into different parts: data cleaning, finding frequent patterns, and interpreting maps.

III. Our Visual Analytics Solution

A. Data Cleaning

As an integral part of dataset preprocessing, data cleaning is a meticulous process dedicated to rectifying or eliminating inaccuracies, inconsistencies, duplications, or gaps in data [33]. The primary aim is to refine and enhance the overall quality of the dataset, rendering it more suitable for subsequent analysis and visualization. This crucial procedure is particularly important as it sets the stage for meaningful insights to be extracted from the data.

During the preprocessing phase, an emphasis is placed on removing data elements that do not contribute to the substantive analysis or visualization of interest. An example within our dataset involves the "INVAGE" column, which may include entries such as "unknown" when attempting to discern the age range of individuals involved in an accident. Given that these "unknown" values lack relevance to our analytical objectives and are incongruent with our visualization goals, the column was not applied.

Algorithm 1 Data Cleaning – Remove "unknown"

```
Input: accidents DataFrame
Output: column without "unknown"
df = accident[["INVAGE", " ALCOHOL"]]
df_clean = df [df ["INVAGE" != "unknown"]]
df_clean
```

To systematically address this issue, we leverage the capabilities of the Python pandas library. By using the functionality provided by pandas, we can efficiently identify and eliminate these extraneous and meaningless entries from the dataset. This data cleaning process ensures that our subsequent visualization efforts are focused exclusively on the meaningful aspects of the data, enhancing the overall integrity and utility of the dataset for analytical purposes.

B. Pattern Analysis Approach

With a large dataset—like the Toronto motor vehicle accident dataset—that has many columns, there are an endless number of possibilities for visualization. To narrow our focus to fewer topics, we chose to apply frequent pattern mining algorithm (e.g., FP-growth algorithm). We had to be very selective in the columns we used because some columns had only a couple of different values and therefore did not contribute in any meaningful way. Thus, we selected the following columns for the frequent pattern mining algorithm:

- LIGHT – Weather is daylight or dark
- NEIGHBORHOOD_158 - # of the neighborhood
- RDSFCOND – Road surface condition (Dry, Wet, ...)
- INJURY – None, Major, Minor
- ACCLASS – Non-fatal, Fatal
- VEHTYPE – Vehicle Type (Automobile, Truck, ...)
- STREET1 – Street Name
- YEAR – year of accident
- TRAFFCTL – Traffic Control (No Control, Traffic Signal)

We set the minsup to 40% and minconf to 85%. The following are the 5 results:

- [{'Daylight'} → {'Dry'}, 0.869]
- [{'Dry', 'Daylight'} → {'Non-Fatal Injury'}, 0.875]
- [{'Daylight', 'Non-Fatal Injury'} → {'Dry'}, 0.871]
- [{'Daylight'} → {'Non-Fatal Injury'}, 0.872]
- [{'Dry'} → {'Non-Fatal Injury'}, 0.858]

We were surprised to see these results as we had expected more accidents when it was dark and wet. However, this gives us insights into the nature of many accidents in Toronto.

C. Data Visualization Process

Prior to generating our maps, we recognized the significance of validating our interpretations through the creation of plots and charts. This step was crucial in ensuring the robustness and reliability of our map-based analyses. While the procedures for generating plots are relatively uniform across all categories, we will highlight key aspects for clarity.

1. To construct the *yearly plot*, we leveraged the existing "year" column within the dataset. Utilizing R, we employed the table function that counts the number of accidents per year. Then, the plotting was done with ggplot.
2. For the *hourly plot*, the dataset conveniently included a time column in the 24-hour clock format (e.g., 6pm recorded as 1800). To enhance data presentation, a new "hour" column was created. We defined time as "t" for this column, divided it by 100, and utilized the floor function to obtain a final number within the range of 0-23 (i.e., Hour = floor(t/100.0)). This adjustment allowed for a broader time frame for accident counting, avoiding the granularity of minute-level data. Subsequently,

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employing R once again, we applied the table function and employed ggplot to generate an hourly plot.

These visualization techniques not only contribute to the clarity of our analyses but also serve as integral steps in refining our subsequent mapping.

D. Map Generation

The Toronto accident dataset contains precise coordinate data detailing incident locations. This column was further divided into two columns with one for longitude and the other for latitude. Additionally, we created another column that contained the frequency of each coordinate.

Our maps have two colour dots based off day and night; thus, we must define day and night before we can continue. Traditionally, daytime is defined as the period when the sun is above the horizon, however, considering the variability of daylight duration across seasons, a consistent definition becomes paramount for our study, necessitating a uniform approach throughout the year.

To maintain uniformity in our analysis, we adopt a standardized definition:

- Day: 06:00 to 17:59
- Night: 18:00 to 05:59

This standardized definition serves as the basis for colour-coded mapping, enhancing the interpretability and reliability of our visualizations.

Mapping all accidents (day vs night) by start mapping with a depiction of all accidents categorized by day and night, coordinates and their frequencies were utilized to generate dots on the map. Dot size was determined according to frequency, adhering to a predefined size ratio—5 for the smallest dot and 25 for the largest. Consideration of accident recency was incorporated by dividing the recorded year range (2006-2022) into four alpha values. These alpha values corresponded to distinct year ranges, with higher alpha denoting more recent accidents.

E. Interpreting Maps

Our primary focus centers on alcohol-related accidents. To enhance our understanding of the maps, we examined general accident trends in the Toronto area. Examination of the charts produced through our methodology indicates a consistent downward trend in the annual accident rate.

IV. Evaluation

To evaluate our visual analytics solution, we applied it to motor vehicle accidents data from a Canadian city of Toronto. Fig. 1 below demonstrates all accidents between 2006 and 2022. As can be observed, there is a significant concentration of dots towards the bottom-middle region, which is the downtown area.

Fig. 2 is of the downtown area only. Even though the density of accidents is prevalent, it can be seen how our method of visualization can pinpoint accident prone areas by noting the larger dots on the map. It is also noticeable that many of the dots are of darker or higher alpha values, which implies the accidents happened recently. These areas present favorable opportunities for city officials to allocate and prioritize public resources effectively.

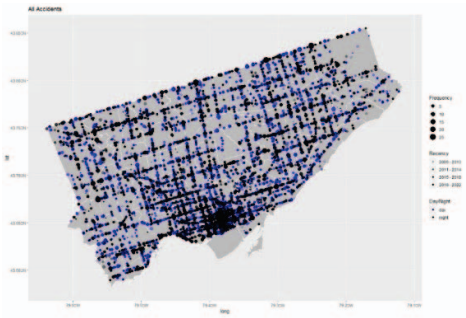

Fig. 1. Motor vehicle accidents in Toronto between 2006 and 2022.

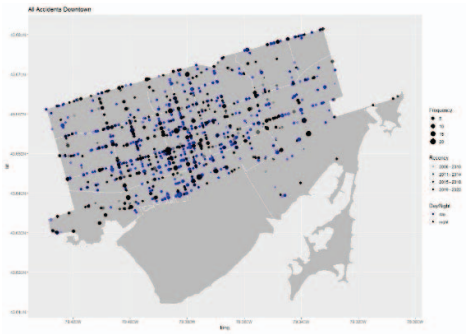

Fig. 2. Motor vehicle accidents in downtown Toronto between 2006 and 2022.

Fig. 3. Alochol-related motor vehicle accidents between 2006 and 2022.

Fig. 3 is a map of all the Alcohol related accidents. The dot size in this map is based upon the frequency of accidents at a given location, ranging from 5 to 25. Like the previous approach, the alpha value was determined by recency, providing insight into the distribution of alcohol-related accidents.

General accident trends in the Toronto area as can be seen in Fig. 4. Examination of the charts produced through our methodology indicates a consistent downward trend in the annual accident rate. It is noteworthy to mention that the data during the COVID-19 period, marked by a lockdown of the entire city, should be disregarded as it significantly influenced and distorted the rates during that time.

Notably, accidents are more frequent in the latter half of the year as can be seen in Fig. 5. The second-half surge may be linked to heightened activity, such as vacation visits and

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increased school-related travel. In addition, there is a subtle uptick on Fridays which can be attributed to increased end-of-week activities as can be seen in Fig. 6.

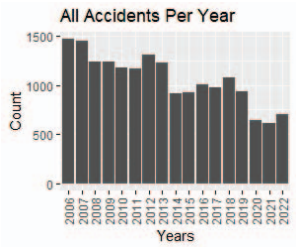

Fig. 4. Accidents grouped by year.

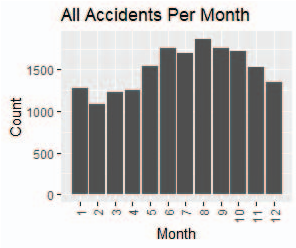

Fig. 5. Accidents grouped by month.

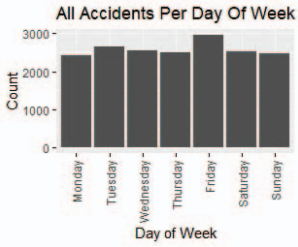

Fig. 6. Accidents grouped by day of the week.

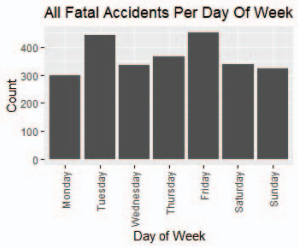

Fig. 7. Fatal accidents grouped by day of the week.

From Fig. 7 a curious observation emerges when analyzing fatal accidents per day of the week, revealing an increase not only on Fridays but also on Tuesdays. Initial searches yielded no clear explanation for the Tuesday surge, necessitating further investigation. Weather events on Tuesdays are posited as a potential cause, but without a deep dive into the exact causes of the accidents, we cannot confirm this theory. Another potential cause are special promotions that occur on Tuesdays such as reduced movie tickets.

The all-accidents map in Fig. 1 has noticeably concentrated incident clusters in downtown Toronto. The submap for the downtown area in Fig. 8 indicates a fairly even distribution of accidents between day and night. Initial predictions suggest heightened daytime traffic attributed to the presence people commuting to and from work and worksites. Bars, clubs, and other nighttime activities can potentially explain elevated nighttime accidents.

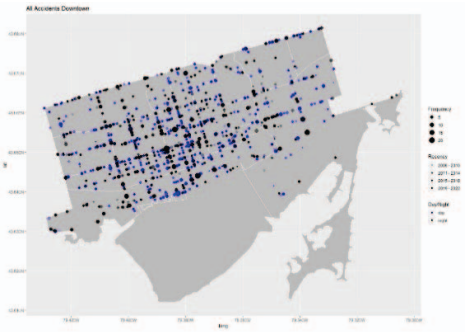

Fig. 8. Distribution of accidents between day and nights.

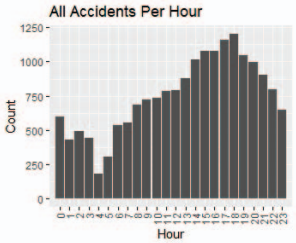

Fig. 9. Distribution of accidents by hours in a day.

Fig. 9 shows all accidents per hour and indicates a skew toward later in the day, aligning with increased evening commute activity and busier roads. As many occupations do not require early morning commutes this aligns with initial predictions. Curiously, the alpha values of dots for both day and nighttime indicate ongoing issues at specific areas with a few lower alpha dots surrounding. Given the pattern of the dots and the alpha values we can predict the locations of intersections. The larger size of the dots in this case suggests these intersections are large and multi-lane. Visual inspection of the intersection at 43.79°N and 79.42°W confirms this theory. This intersection looks to be an issue for a multitude of reasons, including but not limited to multiple lanes, pedestrians crossing areas being close to the traffic stop line, and the presence of exits for parking lots and gas stations. These exits are particularly important because they explain the presence of lower alpha dots surrounding this area of concern.

Transitioning to alcohol-related accidents, a consistent downward trend over the years, as shown in Fig. 10, aligns with overall accident trends indicative of heightened awareness, preventative measures, and potentially improved law enforcement. However, fatal alcohol-related accidents exhibit erratic patterns, with a dip between 2019 and 2021, potentially linked to pandemic-induced closures of restaurants and bars, as well as no fatal accidents occurring during 2011. Unfortunately, 2022 has a high percentage of fatal accidents, which is alarming and cause for further study. One thought is that it may be linked

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to pandemic-induced feelings of loneliness resulting in careless behaviour after being allowed to socialize again.

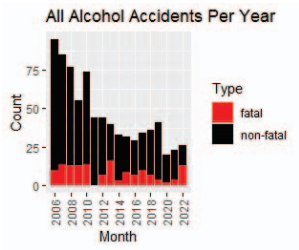

Fig. 10. Alcohol-related accidents by year.

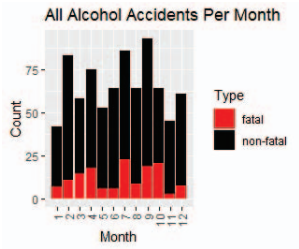

Fig. 11. Alcohol-related accidents by month.

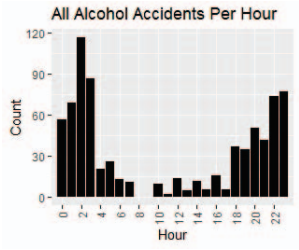

Fig. 12. Alcohol-related accidents by hour.

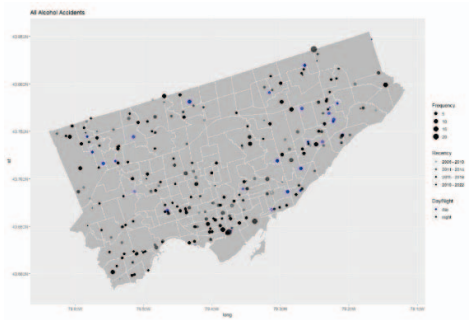

Fig. 13. Alcohol-related accidents by location.

Monthly alcohol-related accidents show no discernible trend, as seen in Fig. 11, with fluctuations seemingly random. However, summer overall does tend to have more alcohol accidents, which is likely due to more parties with friends.

Notably in Fig. 12, alcohol accidents per hour align with expected patterns of alcohol consumption, peaking at 02:00, possibly reflecting post-drinking travel patterns.

Fig. 13, the all alcohol map, corroborates the hourly alcohol chart. It emphasizes nighttime incidents and mirrors overall accident patterns with a heightened concentration in the downtown area that can be attributed to the greater number of bars in the downtown area. There are additional concentrations of nighttime accidents that occur outside of the downtown area. Concentrations in other areas hint at larger traffic volumes and possibly higher average occupant age. The University of Toronto is one such area matching these characteristics. We also expect an area such as the University of Toronto to have a mixture of high and low alpha dots as well as a larger dot size on average. This is because we expect alcohol-related accidents to be an ongoing problem for the area. This can easily be confirmed by a visual inspection of the University of Toronto location on a map.

Fatal alcohol-related accidents lack a discernible pattern, consistent with the myriad factors influencing accident outcomes. Unpredictable elements such as speed, collision angle, and occupant safety measures—e.g., seat belts, all potentially contributing to the variability in accident severity.

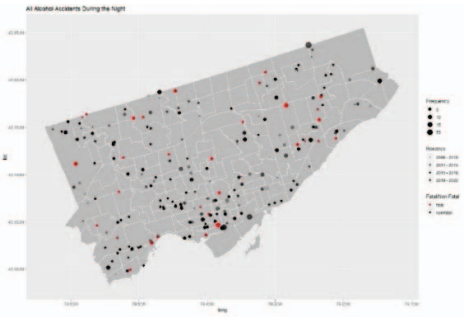

Fig. 14. Alcohol-related fatal accidents by location.

V. Conclusions

In this paper, we presented a visual analytics solution for analyzing motor vehicle accidents in metropolitan cities. To evaluate our solution, we applied it to real-world data in the Canadian metropolitan city of Toronto. Its 158 neighborhoods pose a significant hurdle, making it challenging to visualize accidents in detail. Selecting appropriate colors for contrast is critical to ensure the clarity of our visualizations, especially when dealing with dense points on a map. The complexity of data cleaning adds an additional layer of challenge, requiring attention to detail such as taking into consideration the specifics of the city that may result in relations that do not make sense on initial viewing. With an aim is to visualize all accidents, we separate the data into specific subsets, such as those involving alcohol. Because of varying frequencies and the sheer volume of data, careful consideration in improving legends and enhancing contrast for recency is needed. Moreover, the maps, while informative, may not provide the "why" behind each accident, necessitating additional efforts to look up locations for areas of city service improvement. Evaluation results on real-world data in the Canadian metropolitan city of Toronto show the practicality of our solution in revealing accident patterns in the city. The discovered information and knowledge helps decision makers to improve city services. *Ongoing and future work* could involve incorporating user-friendly features to better interpret the nuances within Toronto's diverse accident dataset

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with things such as submaps and interactive accident lookup features for dots on the map.

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